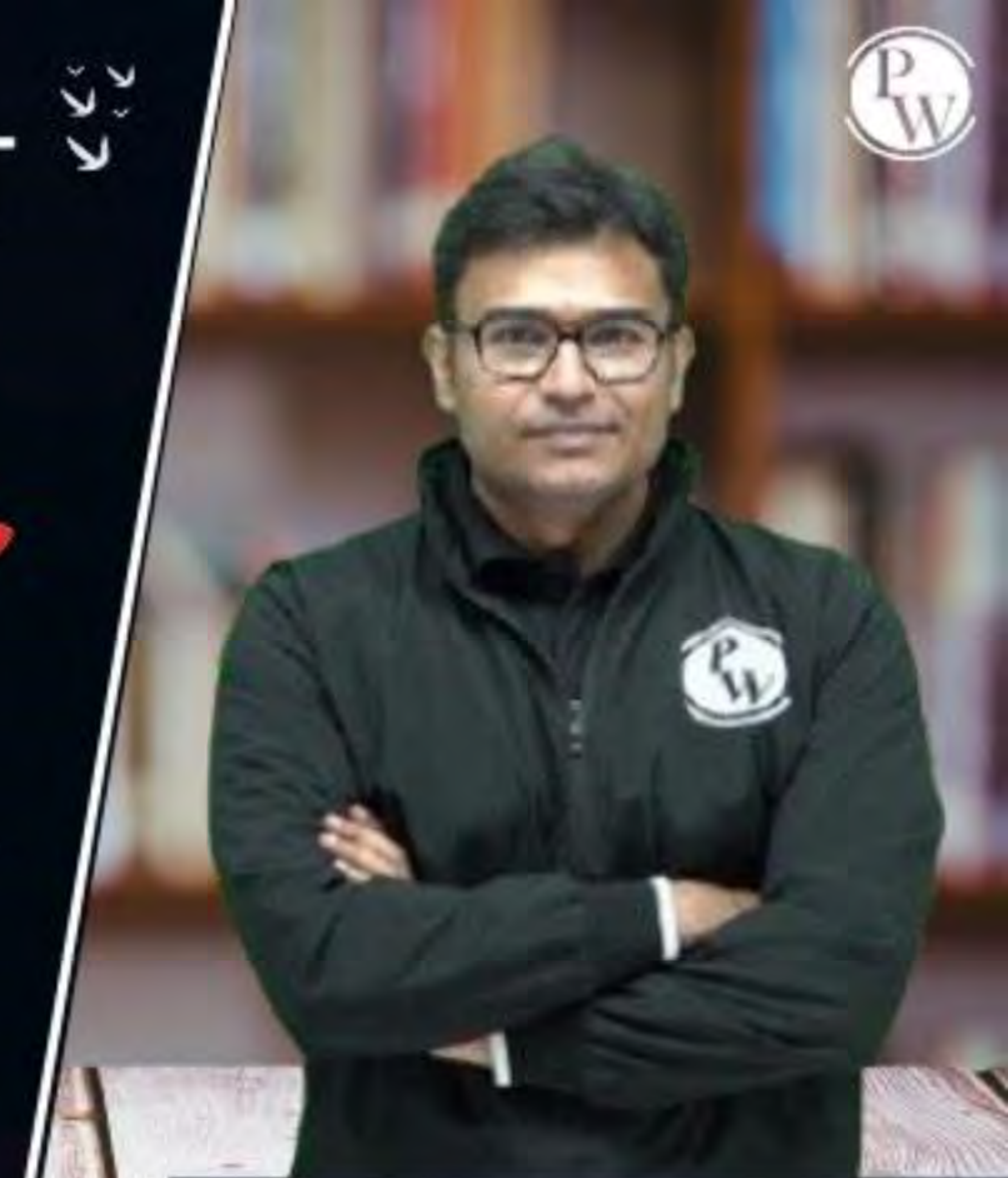
Computer Science & IT

Theory of Computation

Regular Languages

Lecture No. 9



By- Ravindra Sir

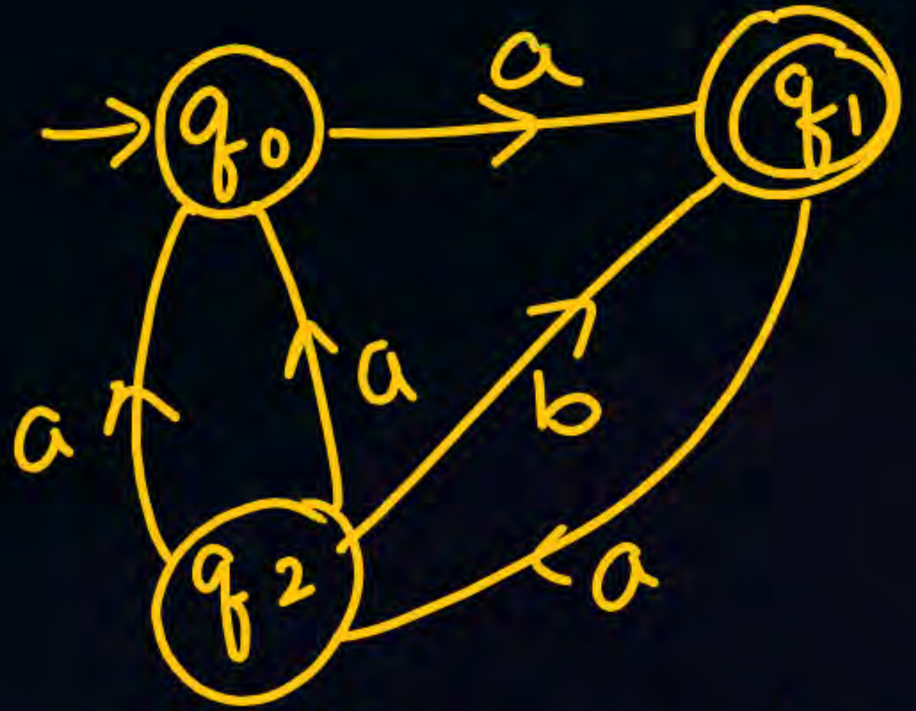
Topics to be Covered

Topic

Topic

Topic

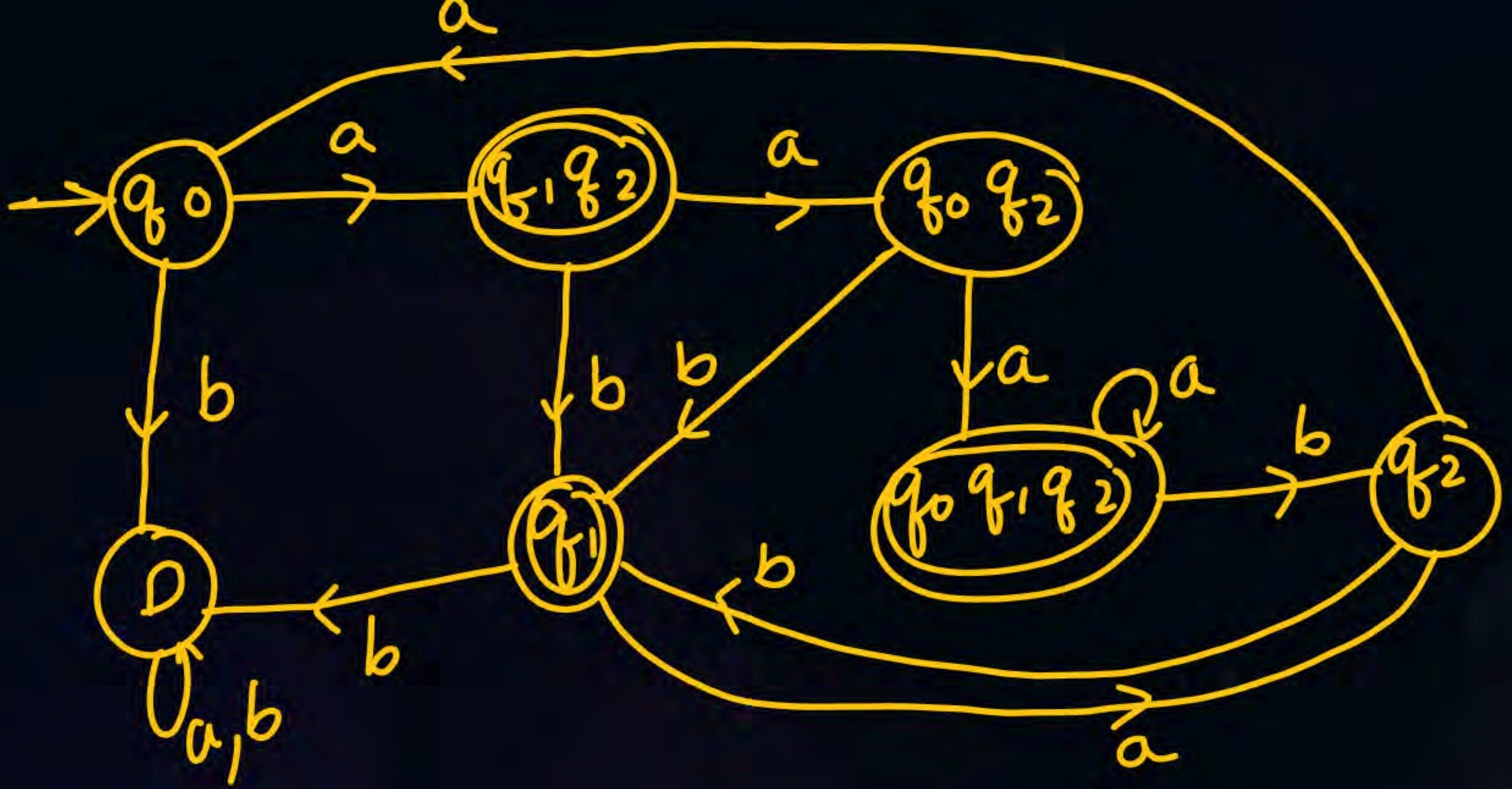
Regular Languages



Transition table

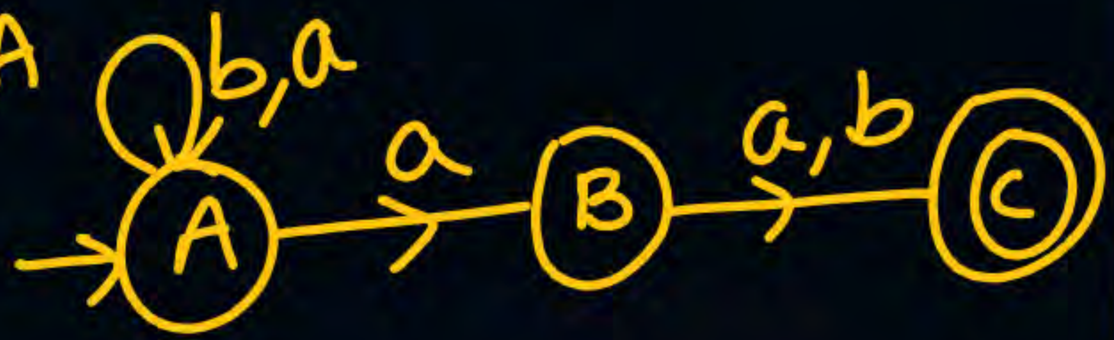
$q_0 \xrightarrow{a} q_1, q_2$

$q_0 \xrightarrow{b}$ Dead


q0
q1 q2
q0 q2
D
q1
q0 q1 q2
q2
a
b
a, b

Second Symbol from RHS is a :- min NFA → 3 states
min DFA → 4 states

NFA



	a	b
A	A,B	A
B	C	C
C	-	-

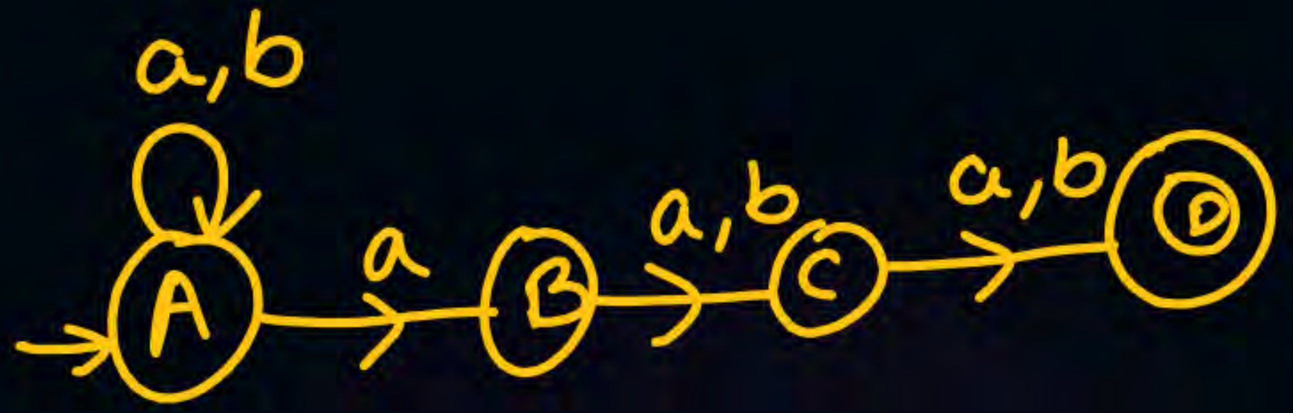
NFA → DFA

may be min	may not be min ✓

	a	b
→[A]	[AB]	[A]
[AB]	[ABC]	[AC]
*[AC]	[AB]	[A]
*[ABC]	[ABC]	[AC]

→ min DFA

III Symbol from RHS is a :



5 min time

min NFA → 4 states
min DFA → 8 states

	a	b
A	A, B	A
B	C	C
C	D	D
*D	–	–

	a	b
→[A]	[AB]	[A]
[AB]	[ABC]	[AC]
[AC]	[ABD]	[AD]
[ABC]	[ABCD]	[ACD]
*[ABD]	[ABC]	[AC]
*[ACD]	[ABD]	[AD]
*[AD]	[AB]	[A]
*[ABCD]	[ABCD]	[ACD]

8 states → min DFA

4th symbol from RHS is a

min NFA → 5 states

min DFA → 16 states

When min NFA has $\underline{O(n)}$ states, then in worst case min DFA can have $\underline{O(2^n)}$ states

2 times gate

DFA $\rightarrow$ min DFA

- If a DFA has n states, then min DFA $\leq n$
- Two states are equivalent if they both either go to a FS or a non FS on all strings
- States A, B are equivalent if $\forall w \in \Sigma^*$ $\delta^*(A, w)$ & $\delta^*(B, w)$ both go to either final state or non final state

$\delta(A, a)$ $A \xrightarrow{a}$ $\delta^*(A, w)$

$A \xrightarrow{w}$

→ A, B are K equivalent, if $\forall\, w \in \Sigma^*$, $|w| \le K$

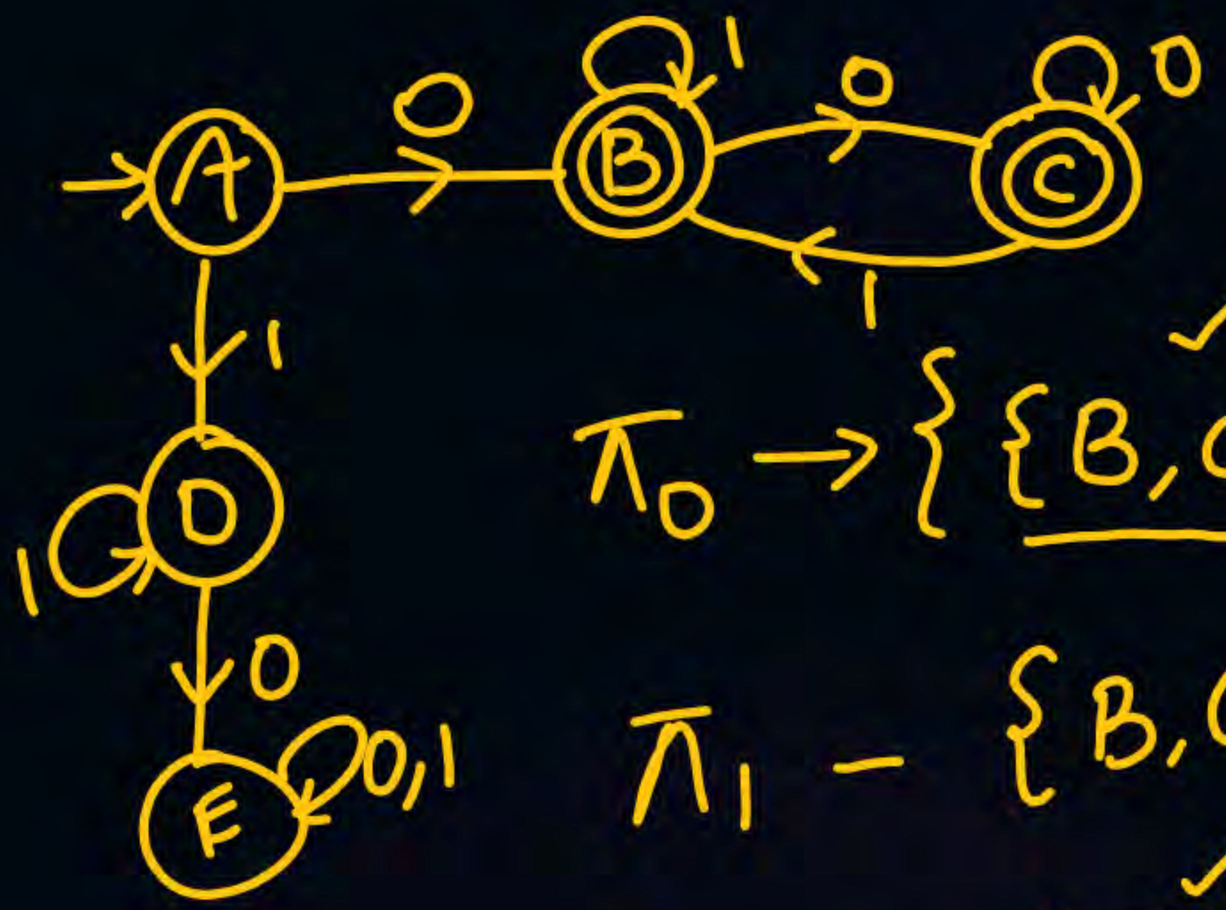
$\delta^*(A, w)$ $\delta^*(B, w)$ both go to either final

State $\wedge$ non final state

0

1

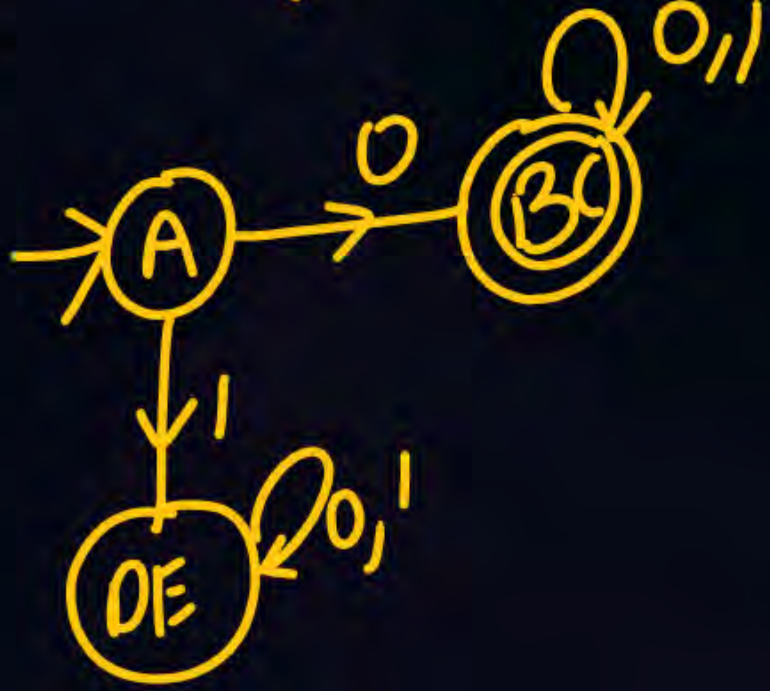
$\Big[\begin{matrix}2\\3\end{matrix}$



$\pi_0 \rightarrow \{\{B,C\}\ \{A,D,E\}\}$

$\pi_1 - \{B,C\}\ \{A\}\ \{D\ E\}$

$\pi_2 - \{B,C\}\ \{A\}\ \{D,E\}$



ϵ-NFA to NFA :-

Identifying regular languages :-

1) If a language is finite then it is regular

$a^n b^n / n \leq 10^{10} \rightarrow$ finite $\rightarrow$ Regular

$a^n b^n c^n / n < 100 \rightarrow$ finite $\rightarrow$ Regular

$ww^R / w \in (a,b)^* \; |w| = 10 \rightarrow$ finite $\rightarrow$ Reg

2) Infinite language →

a) b) c) d)

2) a) Non linear power $\rightarrow$ not regular

$$\left.\begin{array}{l} a^{n^2} / n \geq 0 \\ a^{n!} / n \geq 0 \\ a^{3n^2} / n \geq 0 \\ a^{2n} b^{m^2} / n \geq 0 \\ a^n / n \text{ is prime} \\ a^n / n \text{ is perfect square} \end{array}\right\}$$

Not regular

$$\left.\begin{array}{l} a^n / n \geq 0 \\ a^{2n+3} / n \geq 0 \\ a^{kn+l} / n \geq 0 \\ a^{m+1} b^{n+2} / n, m \geq 0 \\ a^{2m+3} b^{4n+5} / n, m \geq 0 \end{array}\right\} \text{Reg}$$

$\boxed{ax+b}$

2) b) $+, -$ $\times$ $\div$

CFL

CSL

$a^m b^n c^p \mid m+n=p$ $\rightarrow$ ✗ Reg ✓

$m,n,p \geq 0$ ✓ CFL

$a^m b^n c^p \mid m \times n = p$ $\rightarrow$ ✗ Reg

$n,m,p \geq 0$ ✗ CFL

✓ CSL

$a^m b^n c^p \,/\, m+n+p=3 \longrightarrow$ finite

$m, n, p \geq 0$

$\downarrow$

Regular

m	n	p
0	0	3
3	0	0
0	3	0
1	1	1
1	2	0

finite

$a^m b^n c^p \,/\, \frac{m}{n} = p$

$n, m, p \geq 1$

$\times$RL $\times$CFL $\checkmark$CSL

$a^n b^m \,/\, m+n=10 \rightarrow$ finite Regular

$a^m b^n / m+n=p,\ m,n,p \geq 0$

× Reg

✓ CFL

✓ CSL

$a^m b^n \,/\, m+n = 10$

$m+n \neq 10$

$m+n < 10$

$m+n \leq 10$

$m+n > 10$

$m+n \geq 10$

finite $\rightarrow$ Regular

Regular

$L \rightarrow \overline{L}$

$\overline{L} \rightarrow L$

$\overline{a^m b^n / m+n \neq 10} = a^m b^n / m+n = 10$

$\cup$

$(a+b)^* ba (a+b)^*$

$$\overline{a^m b^n / m+n > 10} = a^m b^n / m+n \leq 10 \cup (a+b)^* ba (a+b)^*$$

$$a^{n^2} / n \leq 10^{10} \rightarrow \text{finite langg} \rightarrow \text{Regular}$$

$a^m b^n \,/\, 2m+3n=10$

$\downarrow$

finite $\rightarrow$ Reg

$a^m b^n \,/\, 5m+6n=1000 \rightarrow$ finite

Regular

$a^m b^n / m-n=10$ ✓

∞ lang

$m = n+10 \rightarrow$ CFL

Comparison

Reg

(Reg) ✗

$a^m b^n \mid m-n=10$

$m-n \le 10$

$m-n < 10$

$m-n > 10$

$m-n \ge 10$

$m-n \ne 10$

Non Reg

CFL ✓

$a^m b^n \,/\, m \times n = 10$

$m \times n < 10$

$m \times n \leq 10$

$\rightarrow$ finite Regular

$m \times n > 10$

$m \times n \geq 10$

$m \times n \neq 10$

$\rightarrow \infty \rightarrow$ Regular

$a^m b^n \mid m/n = 10$

$m = 10n$

$\leq$

$<$

$\geq$

$>$

$\neq$

RegX

PDA

CFL

CSL ✓

2) c) <u>Compairision</u> :- RL $\rightarrow$ 0 Compairision

CFL $\rightarrow$ 1 Comparision at a time

CSL $\rightarrow$ any no of Compairisions

$a^m b^n / m, n \geq 0$ $\quad a^* b^* \rightarrow$ Reg

$\rightarrow$ 0 ~~Com~~ Comp

$$\underbrace{a^{m+2}\, b^{3m+4}}_{\text{one comparision}} \;/\; m \geq 0$$

× Reg

✓ CFL

✓ CSL

$a^{m+2} b^{3n+4} / n, m \geq 0$

0 Comp

linear power

$\downarrow$

Reg

$a^{n^2} b^{3n+2}$

Comp → × Reg

× CFL ✓ CSL

one Comp

$$\left\{\begin{array}{l} \overline{a^n b^n} / n \geq 0 \\ a^m b^n / n = m \ \& \ n, m \geq 0 \end{array}\right.$$

× Reg

✓ CFL

$$\left\{\begin{array}{l} a^n \overbrace{b^n c^n} / n \geq 0 \\ a^m b^n c^p / m = n \text{ and } n = p \\ a^m b^n c^p / m = n = p \end{array}\right.$$

× Reg   × CFL   ✓ CSL

$a^m b^n c^p \ / \ m=n \wedge n=p$

×Reg

NDCFL

CFL

NPDA


CSL
CFL
DCFL
Reg

$a^m b^m c^n d^n \,/\, n, m \geq 0$

one comp at a time

✓CFL ✓PDA

✗ Reg

✓DCFL
✓DPDA

✗

✓
CFL

2) d) String matching:-

ww^R, $w \in \Sigma^*$

Even length palindrome

× Reg

a bbbba

Regular ×

CFL → later

$\{ww \mid w \in (a,b)^*\}$

× Reg

× CFL

✓ CSL ✓

aba aba

abb abb

b

b

a

$ww^R / w \in \Sigma^*, |w| \leq 10$

finite

Reg

CFL ✓

WW^R, $|W| = 10$ → finite → Reg

< 10

≤ 10

> 10 → infinite → ✗ Reg

≥ 10 → CFL ✓

$\neq 10$

$\underline{w}\,\underline{w}^R\,\underline{w} / w \in \Sigma^*$

× Reg

× CFL

✓ CSL $\underline{w\,w^R}\;\underline{w}$

$\underline{w_1}\,\underline{w_1}^R\,\underline{w_2}\,\underline{w_2}^R \,/\, w_1, w_2 \in \Sigma^*$

✗ Reg

✓ CFL ✓

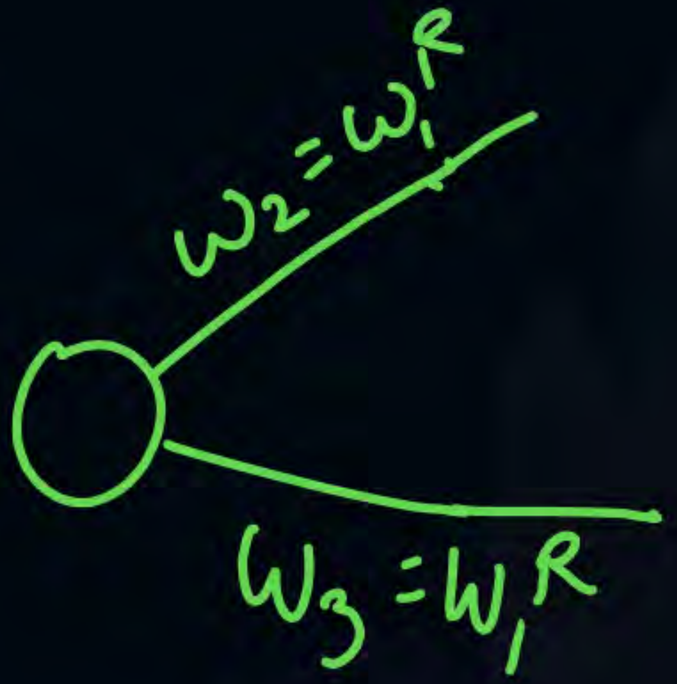
$w_1 w_2 w_3 w_4 \mid w_1, w_2, w_3, w_4 \in \Sigma^*$

$w_2 = w_1^R$ or $w_3 = w_1^R$

NDPA ✓

FA × Reg ×

NPDA ✓



Pumping lemma for Regular languages:-

↳ negative test

$L_1 \rightarrow$ Pass L_1 < may be Reg / may not be Reg

$L_2 \rightarrow$ x

PL

L_2 is not reg

$a^n b^n \rightarrow$

a|aa|a|a b|b|b b|b

a aaaaaa bbbbbb

PL for Reg languages :-

Let L be an infinite reg lang. Then there exists some positive integer m such that $w \in L$ with $|w| \geq m$ can be decomposed as $w = xyz$ with $|xy| \leq m$ and $|y| \geq 1$ such that $w_i = xy^iz$ is also in L for all $i = 0, 1, 2 \ldots$

$a^* \longrightarrow$ $\overset{x}{a\,a}\,|\,\overset{y}{a}\,|\,\overset{z}{a\,a}$ $\underline{a\,a\,a\,a\,a\,a}$

Non reg a^nb^n ✗ $a\,a\,|a|\,a\,|\,b\,b\,b\,b$

$\underline{\underline{abab}}$

Myhill - Nerode theorem:-

↳ Based on min DFA

THANK - YOU